



Birla Institute of Technology & Science, Pilani

Pilani | Dubai | Goa | Hyderabad

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

First Semester 2025-26

Course Handout Part II

Date: 08 January 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number : EEE F111

Course Title : ELECTRICAL SCIENCES

Instructor-in-charge : PRAVEEN KUMAR A.V. (praveen.kumar@pilani.bits-pilani.ac.in, Chamber 2220-R)

[Team of Instructors] : Navneet Gupta, Satyendra Mourya, Praveen Kumar AV, Abhijit Asati, Syed M Zafaruddin, Neeraj Mishra, Bhavesh Chauhan, Dinesh Rano, Pankaj Arora and Karunesh Kr Gupta

1. **Course description:** The course covers basic passive circuit elements, dependent and independent sources, network theorems, circuit analysis techniques, and response of first and second-order circuits. Introduction to three-phase circuits, magnetic circuits, transformers, and the basics of rotating machines. Semiconductors - operation of PN junction diodes, Zener diodes, bipolar junction transistors and field effect transistors. Biasing techniques and applications of diodes and transistors. Introduction to operational amplifiers and applications. Introduction to Digital Electronics
2. **Scope and Objective:** The primary objective of this course is to teach the principles of three different aspects of electrical sciences: (1) Circuit Analysis (2) Electronic Devices and (3) Electromagnetics, to the students of mixed disciplines.
3. **Course learning outcomes:**
 - (i) Students will learn the circuit analysis (voltage, current and power calculation) techniques for DC and AC (1-phase and 3-phase) circuits using Kirchhoff's current law (KCL), Kirchhoff's voltage law (KVL), Ohms law, Nodal analysis, Mesh analysis, Network theorems (Thevenin's theorem, Norton's theorem, Maximum power transfer theorem, Superposition theorem) etc.
 - (ii) Students will be able to perform circuit analysis on electronic circuit consisting of elements such as diode, transistor (BJT, JFET, MOSFET), OPAMP and digital logic.
 - (iii) Students will learn the construction and operation of electromagnetic devices such as transformers and machines



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4. Text books and references:

TB1. Leonard S. Bobrow and Navneet Gupta, Foundations of Electrical Engineering, Oxford University Press, Asian Edition, 2015

RB1. Allan R Hambley, Electrical Engineering: Principles and Applications; 7th Edition (global edition), Pearson education limited, 2019.

5. Course Format & Weekly Workflow

Weekly course work involves 3 lecture hours and 1 tutorial hour.

6. Course Plan:

Week No.	Module	Lec.	Coverage	Ref. (TB)	Learning Outcome
1	Basic electrical circuit laws and elements	1-2	Voltage and current sources, basic circuit components, Kirchhoff's current law (KCL), Kirchhoff's voltage, law (KVL), Instantaneous power, Inductors, Capacitors	1.1-1.7	Students will learn basic circuit elements and the laws of electrical science to solve basic electrical circuits.
		3	Independent and dependent sources, Source transformation	1.8, 2.4	Students will learn independent and dependent sources, source transformation.
2-3	Principles of circuit analysis	4,5	Nodal analysis, Mesh analysis	2.1-2.3	Students will be able to solve circuits by nodal and mesh analysis.
		6-8	Network theorems (Thevenin's theorem, Norton's theorem, Maximum power transfer theorem, Superposition theorem)	2.4,2.6	Students will be able to solve circuits using network theorems.
3-4	Time domain analysis	9-13	Transient response of first and second order circuits (natural and forced response)	3.2-3.5	Students will be able to perform time domain analysis of first and second order circuits.
5-6	AC analysis	14-15	Time-domain analysis, Phasor representation of alternating quantities, j operator and Phasor algebra, Frequency-domain analysis	4.1-4.3	Students will learn concept of phasor, phasor algebra, frequency domain analysis of AC circuits.
		16-18	AC power, Average power, apparent power and complex power, Power factor corrections	4.4-4.5	Students will learn concepts of power in AC circuits.
		19-20	Three phase circuits (Y and Δ connections)	4.6	Students will learn concepts of three phase circuits (Y and Δ connection).



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7	Frequency response and resonance	21-22	Filters - Low Pass, High Pass and Band Pass), Resonance and Quality factor	5.1-5.2	Students will apply frequency domain analysis to filter circuits and resonant circuits
8-9	Semiconductor Devices - Diodes	23-25	Basics of PN junction, Junction diode, Ideal diode, Applications in rectifier and clipper	6.2, (partly) 6.3,6.4,	Students will study basics of semiconductors, junction diodes and their use in various electronic circuits.
		26-27	Zener diode, Applications in voltage regulator and clipper	6.6	Students will learn applications of Zener diodes in electronic circuits.
10	Bipolar Junction Transistors (BJT)	28-30	Brief operation, V-I characteristics of BJT, Circuit analysis	7.1-7.3	Students will learn the operation and analysis of BJT circuits
11	Field-Effect Transistors (FET)	31-33	Brief operation, V-I characteristics of JFET and MOSFET, Circuit analysis	8.1-8.2	Students will learn the operation and analysis of JFET and MOSFET circuits
11-12	Operational amplifier and Digital Electronics	34-35	Basics of operational amplifier and its application	10.1	Students will learn about op-amp circuits and applications.
13-14	Magnetic circuits and transformers	36-37	Analogy between electrical and magnetic circuits, B-H curves	13.1-13.2 (partly) 13.3, 13.4	Students will learn the fundamental concept of electromechanics, magnetic circuits and transformers.
		38-40	Electromagnetic induction, Lenz's law, Transformers (Ideal)	13.6-13.9	Students will learn basic concepts of Ideal transformer.

7. Evaluation Scheme:

Sr. No.	Evaluation Component	Weightage	Marks (300)	Duration	Date & Time	Nature of Component
1.	Class participation*	5 %	15	*	During Lecture Hour	Announced
2.	Quizzes	20 %	60	15 Min.	During Tutorial Hour	Closed Book
3.	Mid-Sem. Exam	30 %	90	90 Min.	14/03/2026 (AN2)	Open Book
4.	Comprehensive Exam.	45%	135	180 Min.	16/05/2026 (AN)	Closed Book



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*** Class Participation Assessment:** This component evaluates students' attendance and class engagement through short quizzes conducted during lecture hours. This score will be considered only for students with a minimum attendance of 50 %. For a student with attendance less than 50 %, zero marks will be awarded, irrespective of the assessment score secured.

- 8. Minimum performance requirement for a valid grade:** A student should secure 25 % of the average of the class, or 10 % of the course total marks, whichever is lower, to clear the course. A student with a score lower than the above threshold will be awarded an NC report.
- 9. Attendance Policy:** Students are expected to attend all classes and participate in all evaluation components.
- 10. Chamber consultation hours:** Will be announced through Nalanda (<https://nalanda.bits-pilani.ac.in>).
- 11. Makeup Policy:** Makeup is not applicable to quizzes or class participation assessment. For other components, makeup will be granted in cases of extreme emergency, supported by sufficient documents.

Instructor-in-charge, EEE F111

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